Large Language Models in Power Systems toward Cognitive Intelligence: A Literature Review

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ABSTRACT

The rapid evolution of power systems, driven by the integration of renewable energy into infrastructure, has fundamentally reshaped the landscape of grid cognitive operation. Modern power systems are now characterized by distributed energy resources and cyber-physical interactions, introducing unprecedented uncertainty and complexity. Under these emerging conditions, conventional machine learning approaches remain narrow and data-dependent, whereas traditional model-based methods even struggle with scalability, adaptability, and robustness. Recently, the advent of Large Language Models (LLMs) indicates a paradigm shift, offering new capabilities in knowledge reasoning, and extending AI from specific tasks to cognitive functions in future power systems. This survey provides a review of LLMs applications, ranging from renewable energy forecasting, microgrid energy management, load forecasting and management, and electricity market trading. Our contributions are summarized as follows: (1) We present a literature review from conventional machine learning and deep learning to LLM-based frameworks, analyzing their capabilities and limitations across each application domain. (2) We discuss the potential applications via LLM extension techniques (e.g., RAG, PEFT, ICL, MLLMs and multi-agents), and their technical challenges in the power system. (3) We identify and discuss open challenges in developing LLMs for real-world power grid environments, emphasizing the promising trends of LLMs drive AI from task-specific applications to cognitive intelligence. Our work can serve as a useful reference for researchers and practitioners, fostering the ongoing development of safe, robust, and intelligent power systems.

1. Introduction

Today, modern power systems are undergoing a rapid transition towards low-carbon and renewable-integrated architectures. The rise of inverter-based resources (IBRs), energy storage, distributed energy resources (DERs), and sector coupling (e.g., power-to-hydrogen, electric vehicles, and integrated energy systems) has introduced unprecedented uncertainty and cyber–physical interactions [1]. Future grids are evolving from passive and hierarchical systems into highly dynamic, data-rich, and interactive systems. However, traditional model-based methods, such as economic dispatch and state estimation, are increasingly inadequate for dealing with the evolving demands of modern grids. This transformation fundamentally reshapes the control philosophy from engineering-driven design to intelligence-driven operation.

Artificial intelligence (AI) has become a key enabler of intelligent grid operation, initially focused on forecasting tasks such as load demand, wind generation, and solar output [2]. As AI technologies advanced, their role expanded to operational decision support [3], including fault diagnosis [4] and voltage/reactive power control [5]. The integration of deep learning and graph neural networks further enabled AI systems to incorporate physical constraints and network topology [6]. However, most existing AI models are still limited in scope and heavily reliant on data, lacking the cognitive abilities necessary for integration with operator workflows, such as decision-making and real-time control [7]. This limitation hinders AI's potential for decision automation and cognitive analytics in large-scale power systems.

Recent advances in foundation models, particularly Large Language Models (LLMs), signal a paradigm shift in AI's role for complex engineered systems [8]. Unlike traditional task-specific models, LLMs show potential for cross-domain cognition and knowledge reasoning. Their ability to integrate domain-specific instructions and interact through natural language prompting enables flexible integration with external tools, simulators, and optimization algorithms. Moreover, LLMs can process diverse system representations (e.g., text logs, SCADA data, video records) within a unified framework. These capabilities position LLMs as a promising foundation for developing general-purpose intelligence frameworks in power systems, extending AI from task-specific applications to cognitive functions such as knowledge augmentation and constraint-aware decision-making. Consequently, increasing efforts are exploring LLM-powered grid copilots for control room assistance [9], autonomous dispatch agents [10], planning and restoration tasks [11], and hybrid architectures combining LLMs with reinforcement learning or optimization for real-time grid operation [12].

Despite growing interests, the applications of LLMs in power systems are still in its early stages and remains fragmented. As shown in Fig. 1, existing efforts focus on isolated application areas, ranging from system forecasting (e.g., renewable source forecasting, load prediction, and fault diagnosis) [13, 14, 15] to operational decision support

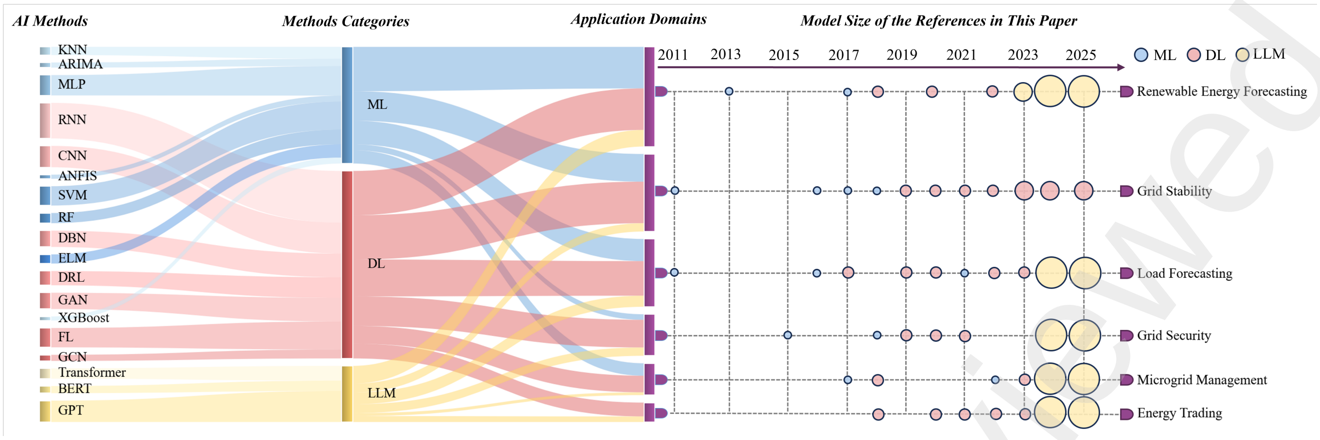

Fig. 1. The proportions of different methods and the annual trends of model sizes in application domains.

(e.g., market advisory tasks, contingency analysis, and grid planning) [12, 16, 17]. While AI systems in power grids have traditionally focused on improving forecasting accuracy, recent advancements in LLMs offer the potential for more complex cognitive operations, such as decision-making and real-time control. However, their application in real-world grid environments is still in the early stages. Key challenges include ensuring the safety and reliability of LLMs within the constraints of grid operations, addressing issues like model verification, reducing hallucinations (i.e., incorrect outputs), and integrating operator supervision. Additionally, the practical integration of LLMs with existing grid infrastructures (e.g., SCADA, EMS, AGC, DMS) remains an ongoing research area, requiring further engineering solutions and methods to build trust in their deployment.

This paper is organized as follows. Section 2 reviews representative surveys on AI for power systems. Section 3 introduces the background of future power systems, LLMs, and key LLM extension techniques that could be applied in power systems. Section 4 surveys renewable-energy forecasting from the perspectives of different time horizons and discusses how LLM-based approaches reshape forecasting pipelines. Section 5 focuses on microgrid engineering, examining LLM-driven energy management and synchronization mechanisms. Section 6 presents load management and load forecasting, emphasizing semantic reasoning, multimodal fusion, and few-shot adaptation. Section 7 explores LLM-based agent systems for electricity markets, including P2P and community trading. Section 9 concludes the paper and outlines open challenges and future research directions.

2. Related Surveys

The growing integration of AI into power systems has motivated a diverse body of survey literatures, broadly focusing on (i) conventional machine learning and big-data analytics for grid operation, (ii) deep-learning methodologies and their applications, (iii) Revisiting market-oriented machine learning with multi-agent systems, and (iv) security and trustworthy AI. While we acknowledge related surveys that cover IoT-enabled smart-grid infrastructures and smart-grid security, our review does not place primary emphasis on these aspects, because we focus on the emerging paradigm of LLMs as a cognitive intelligence layer rather than infrastructure- or protocol-centric issues.

2.1. Conventional Machine Learning and Big Data Analysis for Smart Grid

The IoT infrastructure in smart grids produces massive volumes of heterogeneous data, motivating extensive use of data analytics and conventional machine learning. Zhang *et al.* [18] surveyed big data analytics in smart grids, highlighting applications such as renewable forecasting, load prediction, fault detection, and predictive maintenance. Subsequent studies [19, 20, 21] examined conventional machine learning approaches for tasks including fault diagnosis, maintenance optimization, stability assessment, and forecasting. Although these surveys establish the contribution of conventional machine learning to grid efficiency and reliability, they focus on conventional techniques, with limited coverage of newer paradigms, e.g., reinforcement learning, federated learning, and large foundation models, which are poised to drive future smart-grid intelligence.

2.2. Deep Learning for Smart Grid

Building on the limitations of conventional machine learning, subsequent surveys examined the transformative role of deep learning in power systems. Cheng *et al.* [22] reviewed the evolution of AI from symbolic reasoning to data-driven learning and deep reinforcement learning (DRL), and illustrated the advances in smart-grid capabilities. Omitaomu et al. [23] presented a review that mapped various AI techniques, such as expert systems, fuzzy logic, and supervised learning, to specific grid applications, focusing on their broad applicability rather than methodological innovation. Massaoudi *et al.* [24] surveyed widely used deep learning architectures, including convolutional neural networks (CNNs), recurrent neural networks (RNNs), long short-term memory networks (LSTMs), generative adversarial networks (GANs), and DRL, as well as hybrid designs and enabling technologies such as federated learning and edge intelligence. Powell *et al.* [25] synthesized smart-grid technologies with a focus on cybersecurity, interoperability, and renewable integration, categorizing integration scenarios across distributed generation, large-scale variable resources, and storage-coupled hybrids, and analyzing their operational and economic implications. Zhang *et al.* [26] concentrated on DRL for power systems, outlining its strengths for handling uncertainty and sequential decision-making while noting challenges related to scalability, reward-function design, and reliability. While these surveys provide valuable insights, their focus was primarily on task-specific architectures, with less emphasis on the emerging role of large-scale pre-training and foundation models, which may reshape smart-grid intelligence.

2.3. Revisiting Market-Oriented Machine Learning with Multi-Agent Systems

While conventional machine learning and deep learning are largely concerned with pattern recognition, a different perspective has emerged in the literature, focusing on how agent-based intelligence reshapes decision-making in modern energy markets. In prior surveys, Ghoddusi *et al.* [27] traced the shift from classical optimization to neural networks and support vector machines for risk management. In computational economics, Aggarwal *et al.* [28] showed that machine learning outperforms traditional econometric models (e.g., ARIMA) in capturing nonlinear, high-dimensional market dynamics. Tushar *et al.* [29] surveyed peer-to-peer (P2P) trading and game-theoretic formulations, distinguishing virtual and physical layers for adaptive bidding and prediction-aware auctions. Recently, Shah *et al.* [30] likewise focused on P2P trading, showing that multi-agent systems (MAS) enable autonomous negotiation and blockchain enhances trust, transparency, and settlement in decentralized markets. Expanding the lens, Izmirlioglu *et al.* [31] reviewed MAS across the smart-grid architecture, emphasizing roles in distributed control, self-healing, and microgrid management to enhance resilience. However, these surveys largely center on specialized algorithms and market-oriented coordination frameworks, leaving open a systematic examination of how LLMs can unify perception, reasoning, and control across grid planning and operation.

2.4. Trustworthy and Secure AI for Smart Grids

As smart grids evolve into tightly coupled cyber–physical systems, security remains a foundational requirement. Unlike "best-effort" IT services, power systems operate under strict physical constraints and near-zero tolerance for failure, demanding particular caution when integrating AI. Prior surveys by Goudarzi *et al.* [32] and Tufail *et al.* [33] systematically cataloged infrastructure vulnerabilities across AMI, SCADA, and communication protocols, while Mollah *et al.* [34] examined blockchain as a means to enhance data integrity and privacy, thereby establishing the defensive baseline for grid communication. The rapid adoption of deep learning introduces new algorithmic attack surfaces. Hao *et al.* [35] surveyed adversarial machine learning in smart grids, showing that forecasting and diagnosis models can be undermined by evasion and poisoning attacks. Beyond robustness, trustworthiness is equally critical. Machlev *et al.* [36] reviewed Explainable Artificial Intelligence (XAI) techniques and argued that the "black-box" nature of neural networks limits operator trust, making improved explainability a prerequisite for deploying AI in power systems. While providing valuable insights into cybersecurity, adversarial robustness, and explainability, these surveys focus primarily on conventional AI pipelines and overlook LLM-specific risks, such as hallucinations and prompt-injection attacks. Given the probabilistic nature of LLM generation, it can conflict with the reliability guarantees needed for grid dispatch. Therefore, accuracy alone is insufficient, and LLM-based methods must be designed with worst-case safety.

While the aforementioned surveys have made significant contributions to understanding AI applications in power systems, they predominantly focus on conventional machine learning and deep learning paradigms, treating AI as a task-specific optimization tool rather than as a cognitive intelligence layer. This survey distinguishes itself from prior work in several key aspects. First, we examine the paradigm shift brought by LLMs, shifting from task-specific models to general-purpose intelligence frameworks capable of reasoning, natural language interaction, and cross-domain knowledge transfer. Second, we provide a comprehensive overview of LLM-based methods and their specific applications across renewable energy forecasting, grid stability assessment, microgrid management, load forecasting and management, energy market trading, and cybersecurity scenarios. We also emphasize the practical deployment challenges of LLMs in real grid environments, including real-time performance constraints, domain knowledge integration, safety verification, hallucination mitigation, and regulatory compliance. Finally, we provide future research directions for each LLM extension technique, identifying open challenges such as the incorporation of physical constraints, multimodal alignment, and communication efficiency in multi-agent systems. Table 1 summarizes the key differences between this survey and representative prior work.

3. Preliminary

3.1. Future Power System

The concepts of China's "New-type Power System" [38], the U.S. "Future Power System" [39], and Europe's "Climate-neutral Energy System" [40] share a common vision: mitigating climate change by transitioning from centralized, fossil fuel–dominated grids to distributed, low-carbon, and intelligent infrastructures. This emerging paradigm, known as the smart grid [41], integrates advanced monitoring, control, and communication to improve resilience and service quality.

Table 1
Comparison of This Survey with Representative Prior Surveys

Survey	Deep Learning	Federated Learning	Reinforcement Learning	Large Language Model
Rathor *et al.* [37]	✗	✗	✗	✗
Zhang *et al.* [18]	✗	✗	✗	✗
Cheng *et al.* [22]	✓	✗	✓	✗
Massaoudi *et al.* [24]	✓	✓	✓	✗
Tushar *et al.* [29]	✗	✗	✓	✗
Tufail *et al.* [33]	✓	✗	✗	✗
This Survey	✓	✓	✓	✓

Although the national contexts are different, China emphasizes policy-driven large-scale renewable integration and grid-centric coordination, whereas the United States stresses resilience, market mechanisms, and cyber-physical integration. Nevertheless, both face similar challenges: integrating a rapidly growing share of renewable, distributed generation resources, ensuring reliability under growing complexity, and coordinating multiple stakeholders. As shown in Fig. 2, future systems interconnect diverse resources (wind, solar, fuel cells, and thermal power) with households, commercial users, and industry through an architecture that unifies generation, transmission, consumption, and control. To address these pressures, the systems couple energy and data flows to balance supply and demand in real time, creating a more flexible and intelligent grid.

AI has progressively addressed significant barriers [42] in integrating renewable energy into grid operations. It has improved forecasting, optimized scheduling and control, and enhanced anomaly detection, fault localization, predictive maintenance, and demand-response modeling [43, 44]. With the emergence of foundation models, capabilities are shifting from task-specific optimization to cognitive decision support. LLMs enable natural-language interaction, reasoning, and semantic interpretation of system events [8]. Multi-modal models fuse sensor, weather, and market data for better prediction [45]. Digital twins increasingly integrate generative AI for scenario simulation and risk assessment [46]. Large models further support cross-domain knowledge transfer across electricity, heat, gas, and mobility [47]. AI has evolved from tools that merely cope with variability and complexity to LLM-driven systems that help manage and coordinate future power systems.

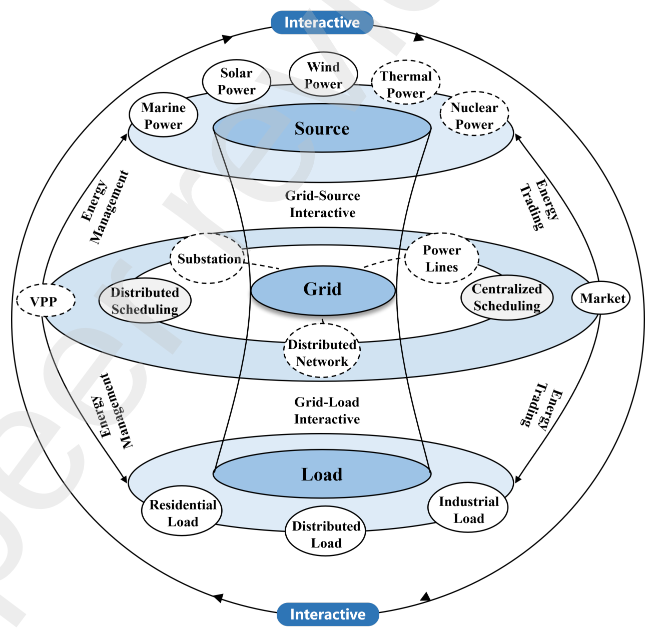

Fig. 2. Architecture diagram of a new power system.

3.2. Large Language Model

In recent decades, deep learning methods, e.g., CNNs [48], RNNs [49], and Transformers [50] have been extensively applied in smart grids[51, 48, 52, 53, 54], supporting tasks including load forecasting, renewable prediction, anomaly detection, and asset monitoring. These models effectively capture spatio–temporal dependencies and typically outperform traditional statistical approaches when sufficient labeled data are available.

However, current power systems introduce challenges that conventional deep learning models are not designed to address. First, grid operations occur in heterogeneous, evolving environments where much of the domain knowledge is encoded not in numerical data, but in text—such as procedures, manuals, incident reports, market rules, and regulations. Task-specific models cannot reason directly over such diverse sources. Second, existing models struggle to integrate physics, rules, and data in a consistent manner. Although they fit historical trends well, they may generate unsafe decisions without principled justifications, which violate protection margins, ramping limits, or operational policies. These limitations hinder deployment in safety-critical settings.

The rise of foundation models shifts AI from a task-specific tool toward cognitive decision support. Since GPT-3, LLMs have demonstrated strong in-context learning and natural-language reasoning [55], and are now being explored for renewable forecasting, microgrid management, load modeling, market trading, and grid security [56, 57, 58, 59]. LLMs offer a new paradigm: they can integrate multimodal, cross-domain, and heterogeneous information while enabling cognitive intelligence. Rather than acting only as predictors, LLMs function as cognitive mediators capable of: (1)unifying textual logs, tabular records, and

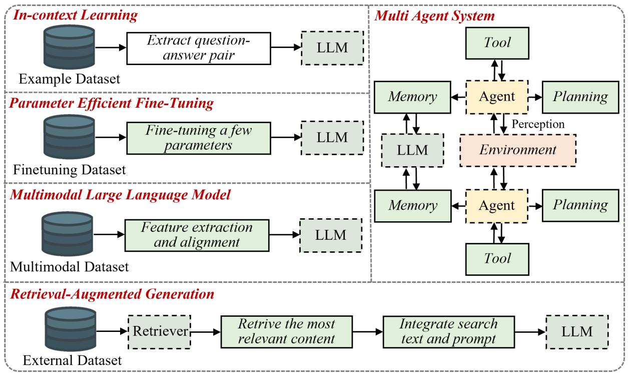

Fig. 3. Extension methods of a large language model.

symbolic procedures into a single reasoning space; (2)following natural-language instructions to coordinate with simulators, optimization solvers, and databases; (3)explaining decisions through interpretable reasoning traces or tool-invocation workflows. However, these capabilities remain largely untested under stringent grid-safety requirements. LLMs are prone to hallucinations, lack formal guarantees, and incur inference latency incompatible with fast control loops. Thus, LLMs are better positioned as decision copilots that orchestrate interactions among physics-based models, optimization frameworks, human operators, and evolving policy constraints.

3.3. Extension methods for Large Language Models

As shown in Fig. 3, a wide range of extension techniques have emerged around LLMs, including Parameter Efficient Fine-Tuning (PEFT) [60], Retrieval-Augmented Generation (RAG) [61], In-Context Learning (ICL) [62], Multimodal Large Language Models (MLLMs) [63], and LLM-based Agent Systems [64]. Rather than considering them as isolated technical add-ons, they are more meaningfully understood as principled ways to bridge persistent gaps between power systems and LLMs. Each technique will be introduced separately, which addresses a different and recurring challenge in real-world deployment.

3.3.1. Parameter Efficient Fine-Tuning

Given real-world limited historical data, e.g., new generation assets, protection strategies, and microgrid topologies, Parameter Efficient Fine-Tuning (PEFT) [60] enables lightweight adapters that specialize a general model to local conditions while preserving global knowledge. This approach requires fine-tuning only a small number of parameters, thereby reducing GPU memory usage and training time. Moreover, considering the diverse operational scenarios in power grids, PEFT allows for the creation of lightweight modules tailored to specific scenarios, eliminating the need to maintain a complete model for each task. This greatly enhances the flexibility of model deployment. In practical applications, PEFT has been successfully implemented in renewable energy forecasting, where it allows the integration of external information to update model knowledge and improve prediction accuracy. As shown in Table 2, PEFT methods are primarily categorized into (1) Prefix-Tuning, (2) Prompt-Tuning, (3) Adapter-Tuning, and (4) Low-Rank Adaptation (LoRA), based on the method and the location of parameter modifications. Specifically, these categories differ in whether they modify input embeddings (Input), attention layers (Attention), or feed-forward layers (FFN). Additionally, some techniques allow for merging into the base model weights after training (Mergeable), while others may introduce additional inference latency (Extra Latency).

3.3.2. Retrieval-Augmented Generation

The power system domain is highly specialized and continuously evolving. First, grid operation involves numerous technical terms, industry standards, historical operation records, and real-time monitoring data. Second, as modern power systems advance, knowledge in areas such as distributed energy integration, power market mechanisms, and intelligent dispatch strategies is frequently updated. Thus, the pre-training corpora of general LLMs often lack domain-specific expertise and cannot easily incorporate the latest industry developments.

Retrieval-Augmented Generation (RAG) [61] technology improves LLM accuracy and reliability by retrieving relevant external knowledge, which could address this limitation by providing domain-specific and up-to-date knowledge support for LLMs. Currently, RAG has been applied in various power system scenarios, including fault analysis assistance systems for smart grids, knowledge-based Q&A platforms for power equipment, and intelligent decision-support tools for energy management systems.

3.3.3. In-Context Learning

The operation and management of power systems are characterized by diverse application scenarios, high demands for real-time decision-making, and limited availability of labeled data. In practice, power grids encounter a wide variety of fault types, each requiring distinct handling strategies. Extreme disturbances and rare failures offer few labeled samples, yet operator practice contains rich procedural insight. In-Context Learning (ICL) [62] enables rapid adaptation through carefully curated examples without parameter updates, which guides LLMs to understand task intent by providing a small number of task-related examples in the input prompt. ICL could help address these challenges by enabling reasoning with only a few historical cases, eliminating the need for extensive retraining.

3.3.4. Multimodal Large Language Model

Modern power grids generate an enormous amount of heterogeneous data, including telemetry streams, satellite imagery, topology maps, alarm logs, and market records. Multimodal Large Language Models (MLLMs) [63] fuse such heterogeneous inputs into a unified representation space that enables cross-modal reasoning. Their architecture typically consists of modality-specific encoders, a cross-modal alignment module, and an LLM backbone: encoders

Table 2
Comparison of PEFT Techniques

Method	Input	Attention	FFN	Mergeable	Extra Latency	Params
Prefix-Tuning	✗	✓	✗	✗	✓	<1%
Prompt-Tuning	✓	✗	✗	✗	✓	≪1%
Adapter-Tuning	✗	✓	✓	✗	✓	1–5%
LoRA	✗	✓	✓	✓	✗	<1%

transform raw inputs into feature vectors, the alignment module maps them into a shared space, and the LLM performs sequence reasoning and generation. Large-scale pretraining endows MLLMs with transferable prior knowledge, enabling faster adaptation to new tasks and reducing the need for extensive manual labeling.

In practice, MLLMs have been explored in renewable energy forecasting, fault diagnosis, and intelligent dispatch-from correlating drone imagery with equipment records to generate automated inspection reports, to integrating topology maps and monitoring data for context-aware recommendations and risk alerts. Designing physics-aware encoders and shared embedding spaces is therefore central to improving reliability and interpretability, ensuring that multimodal reasoning remains consistent with physical grid constraints.

3.3.5. LLM-based Agent Systems

Power systems are inherently distributed and dynamic, with decision authority spread across control layers, assets, and organizations. This makes them naturally suited to multi-agent formulations, where multiple specialized agents handle localized tasks while coordinating their actions through communication mechanisms to maintain system-level objectives. LLM-based Multi-Agent Systems (MAS) [64] extend this paradigm by enabling agents to reason, negotiate, and coordinate using shared representations of the environment. Agents may collaborate, debate, or even compete within sandbox, physical, or information-driven environments, allowing complex behaviors to emerge that are difficult to engineer in single-agent systems. In such frameworks, role definition, environment modeling, and structured communication become central design elements.

In power systems, MAS naturally align with hierarchical control structures. By enabling multi-level collaboration, MAS facilitate strategic planning, operational scheduling, and real-time control, making dispatch more responsive to changing grid conditions. Agents can continuously monitor system states, learn from past actions, and adjust strategies in real time, enhancing resilience to disturbances, such as fluctuations in renewable energy generation. Beyond operations, multi-agent game modeling captures the competitive and cooperative dynamics between generators, retailers, and consumers, supporting the optimization and design of electricity market mechanisms.

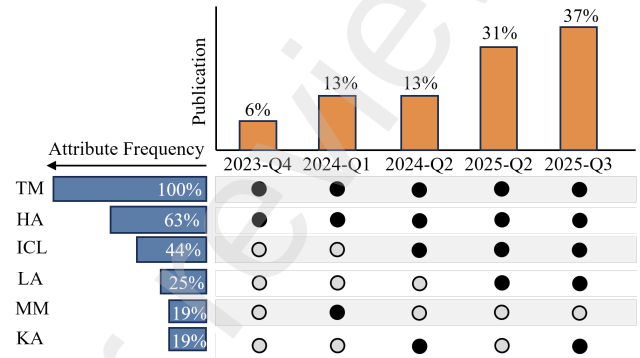

Fig. 4. Distribution of LLM-based renewable energy forecasting studies.

4. Renewable Energy Resources Forecasting

We first summarize the methodological attributes implemented in the reviewed LLM-based forecasting studies. As shown in Fig. 4, the top part shows the publication count by quarter. The left part shows the attribute frequency across all papers. The center part shows attribute presence indicated by filled circles and absence indicated by empty circles across periods. It is noteworthy that 100% of the works incorporate time-series modeling (TM), indicating that LLMs are almost always embedded within conventional forecasting pipelines rather than replacing them. Moreover, 63% combine LLMs with hybrid architectures (HA), whereas 44% exploit in-context learning (ICL). Only 25% investigate lightweight adaptation (LA), and merely 19% address multimodality (MM) or knowledge augmentation (KA). This pattern reveals that most existing studies emphasize model integration and prompt-level learning, while more advanced directions-such as multimodal fusion and explicit knowledge enhancement-remain underexplored. Nevertheless, these attributes appear increasingly in the most recent quarters, suggesting emerging potential for richer, more context-aware LLM-based forecasting frameworks [45, 56]. Renewable generation is highly weather-dependent, and intermittent, posing significant challenges for integration.

Building on these methodological observations, we next examine the forecasting problem itself and its manifestation across different time periods. Forecasting is commonly divided into long-term, short-term, and ultra-short-term horizons. Long-term forecasting requires models that capture long-range temporal dynamics. Conventional machine learning approaches relied on statistical regression using historical generation records, meteorological reanalysis, economic indicators, and geographic features [65, 66, 67, 68, 69], but their reliance on manual feature design and prior assumptions limits generalization under nonstationary and multi-scale conditions. Deep learning methods, including Deep Q-Networks, offer improved adaptability through dynamic strategy selection [70], yet they still suffer from overfitting when long-term samples are scarce. For short-term forecasting, numerical weather predictions are typically combined with historical generation data [66, 71, 72], while end-to-end deep learning improves accuracy by learning spatiotemporal patterns automatically [73, 74, 75, 76, 77]. However, performance remains sensitive to uncertainty and extreme weather. In ultra-short-term scenarios, lightweight machine learning emphasizes fast deployment [78, 79, 80, 81], whereas deep models better capture high-frequency dynamics [82, 83, 84, 85, 86, 87, 88, 89, 90]. Nevertheless, most forecasting models remain strongly dependent on training distributions and often require lengthy retraining when system configurations change.

Although these approaches have improved accuracy and robustness, two key limitations persist: models are typically tailored to specific datasets and operating contexts, and they cannot reason over heterogeneous knowledge sources such as logs, reports, or planning documents. These issues motivate a shift from purely statistical or learning-based predictors toward cognitive models capable of combining numerical data with symbolic and textual information. LLMs offer such potential: instead of acting only as forecasters, they can coordinate tools, interpret contextual knowledge, and adapt forecasting pipelines across time scales. In future grids, forecasting becomes a cognitive input rather than the final objective-LLMs orchestrate models, interpret events, and reshape dispatch workflows. On this basis, we next examine renewable-energy forecasting from the perspective of LLM-enabled methodologies across long-, short-, and ultra-short-term horizons.

4.1. Long-Term LLM-based Forecasting

Currently, the application of large pre-trained models in long-term forecasting is still in the early stages. Existing LLMs, pre-trained on massive text corpora, are capable of understanding implicit relationships between climate, energy, and economic variables. However, they have not yet been applied to long-term forecasting. This gap stems primarily from the modality mismatch: long-term planning relies heavily on numerical simulation reports and geospatial data, whereas current LLMs struggle to ground their semantic reasoning in these precise physical models without extensive multimodal alignment. As discussed in Section 3.3.1 and Section 3.3.2, RAG can incorporate the latest climate models and policy documents, while PEFT can adapt foundation models to specific geographical regions with limited computational resources. Integrating these technologies could enhance predictive capabilities.

4.2. Short-Term LLM-based Forecasting

Recent research on LLMs for short-term forecasting has focused on combining the reasoning capabilities of LLMs with the time-series modeling strengths of traditional forecasting models, creating complementary advantages (e.g., Table 3). Specifically, the LLM serves as a decision-making benchmark, leveraging smaller external models to extend its functional boundaries. For example, Akinci *et al.* [13] proposed a hybrid architecture that combines decision trees and LLMs for daily wind power density classification. Zhang *et al.* [91] integrated wavelet transforms with the LLaMA model, first decomposing photovoltaic time series data into multiple smoothed sub-frequency components using wavelet decomposition, and then utilizing LLaMA's sequence modeling capabilities to generate weekly probabilistic forecasts for each component. He *et al.* [92] proposed LLMGeovec, a training-free method that uses LLMs combined with OpenStreetMap data to derive universal geolocation representations, which boost the performance of forecasting models on solar energy datasets. In these approaches, the zero-shot reasoning abilities of LLMs help address the limitations of traditional models in handling unusual weather patterns. However, the capabilities of large models and traditional models differ significantly. Traditional models may not fully understand the semantic information in the text sequences generated by large models, while large models that lack specific knowledge in the power field may struggle to interpret the outputs of smaller models, potentially leading to deviations in prediction results. Fundamentally, this highlights a "Reasoning-Modeling Gap": traditional models excel at extracting high-frequency numerical patterns but lack the semantic context to interpret low-frequency anomalies (e.g., extreme weather alerts), while LLMs possess the context but lack numerical precision. Therefore, the most promising direction is not to replace traditional predictors, but to use LLMs as a "Meta-Controller" that dynamically selects or weights traditional models based on environmental context. As discussed in Section 3.3.4 and Section 3.3.3, MLLM and ICL techniques show that LLMs can integrate heterogeneous data sources, including numerical weather predictions, satellite imagery, and textual weather reports, into a unified representation space. This enables LLMs to learn knowledge without requiring extensive retraining, demonstrating their powerful adaptability for downstream tasks. Moving forward, more attention could be given to enhancing LLMs' ability to model temporal tasks directly, rather than just serving as a feature extraction tool for smaller models.

4.3. Ultra-Short-Term LLM-based Forecasting

LLMs, pre-trained on massive text corpora, have acquired substantial semantic knowledge [93, 94, 95]. As shown in Table 3, these models can be adapted to new forecasting tasks through lightweight fine-tuning methods, such as SFT and PEFT, using only a small amount of new data without the need to pre-train the entire model [96, 97, 98, 99]. However, due to the large number of parameters in LLMs, methods like PEFT, discussed in Section 3.3, still require substantial computational resources. To address this, techniques such as RAG and ICL allow the injection of external, task-relevant knowledge-such as the latest meteorological observations, expert insights, and physical constraints-into the context window of large models. This enables real-time updates to the model's knowledge base and enhances performance on the current task [100, 101, 102, 103]. Ultra-short-term forecasting involves data from multiple modalities, such as time series, remote sensing imagery, and text descriptions. MLLMs integrate these heterogeneous data sources into a unified representation space. By incorporating prior information from other modalities, the model gains richer environmental perception and improves its ability to handle uncertain data [104, 105, 45, 56]. However, LLMs are primarily trained on text data, which must be tokenized using methods such as byte pair encoding (BPE).Time series data in power systems often consist of floating-point numbers, which can be incorrectly tokenized. This can result in decimal points and adjacent numbers being merged, causing the model to misinterpret the data and fail to learn accurate representations. Moreover, the semantic misalignment between time series and text modalities may further degrade performance. This "Continuous-Discrete Mismatch" remains the central bottleneck for end-to-end LLM forecasting. While standard tokenizers (like BPE) are optimized for text, they fragment time-series continuity, suggesting that future works must develop "Time-Aware Tokenizers" or patch-based embedding strategies to preserve the temporal integrity of power data. The MAS framework presented in Section 3.3.5 offers a promising architecture in which specialized agents handle different modalities. This approach may circumvent the tokenization limitations of single-model architectures while enabling more flexible and accurate ultra-short-term forecasting systems.

Discussion on Future Directions: Current forecasting models face challenges when handling data from new power plants or integrating diverse data modalities. In renewable energy forecasting, historical data is often scarce during the early stages of new power plant commissioning. PEFT enables models trained on similar regions to be efficiently adapted to new sites with minimal data. Similarly, ICL enables models to quickly adapt to new regions using only a few examples from the target area, reducing the costs of knowledge transfer. Regarding data integration, MLLMs combine meteorological satellite cloud images, historical power generation time-series data, and numerical weather forecast text to improve forecast accuracy. However, the primary data in power grids is often represented as continuous floating-point time-series data, while LLMs are pre-trained on discrete tokens. Directly converting numerical values into text tokens leads to semantic fragmentation, making it difficult for the model to understand the continuity and magnitude relationships within the numerical data. To address this, it may be necessary to design specialized tokenizers or continuous embedding layers to directly map floating-point numbers into embedding vectors, while still maintaining floating-point precision.

5. Smart Microgrids

A microgrid is a small-scale power system that can operate in either grid-connected or islanded mode, consisting of distributed generation, energy storage, loads, and control systems, offering a high degree of autonomy (Fig. 5). It operates in a dynamic environment with multiple interacting agents, requiring an efficient energy management system (EMS) and synchronized control mechanism. The EMS coordinates the generation, storage, and load of distributed energy resources to balance supply and demand, while the synchronized control mechanism ensures system frequency and phase consistency, enabling the coordinated operation of distributed generation units. From an optimization perspective, microgrid operation can be modeled as a distributed optimization problem. Before the rise of LLMs, methods such as game theory, mathematical optimization, and distributed learning were commonly used for microgrid scheduling. With the advent of LLMs, the pre-training and fine-tuning paradigm, along with extended techniques like RAG and ICL, has enabled the integration of domain knowledge and adaptation to dynamic scenarios. This section examines how LLM-based techniques are transforming both energy management systems and synchronization mechanisms, focusing on their capabilities in semantic reasoning, dynamic adaptation, and human-AI collaboration for intelligent microgrid operation.

We first investigate the technical attribute distribution of LLM-based smart microgrid studies in Fig. 5. Each row corresponds to a reviewed work and each column to a specific capability of the deployed LLM system. The heatmap reveals that only a small subset of papers implement more than two technical attributes simultaneously, suggesting that current systems are still relatively specialized rather than fully integrated. Semantic reasoning (SR) and optimization integration (OI) appear most frequently, reflecting a predominant use of LLMs for decision support, planning, and interpretation of system states. By contrast, knowledge augmentation (KA), multi-agent system (MAS), real-time adaptation (RTA), and explainability (XAI) occur far less often, and advanced capabilities such as hybrid reinforcement learning (HRL) or autonomous tool and code generation (TC) are present only in isolated cases. Therefore, although LLMs are increasingly used for higher-level reasoning and decision support, most implementations still operate in static or offline modes, rather than interacting with microgrids in real time [58, 14, 106].

5.1. Energy Management System

Recent research on EMS based on LLMs has focused on three main directions. The first direction enhances LLMs with optimization algorithms, by leveraging the reasoning capabilities of large models to support traditional optimization techniques, which could understand operating constraints and optimization objectives through natural language. LLMs contribute semantic understanding by analyzing logs, procedures, and operator intents to guide numerical optimization processes [107, 12], which numerical

Table 3
Large Language Model Approaches in Renewable Energy Forecasting

Ref.	Horizon	Application	Base Model	Pre-trained	PEFT	SFT	ICL/Prompt	RAG	MLLM	API Call
Liu *et al.* [94]	15 min	Wind, Solar	Transformer	✓	✗	✓	✗	✗	✗	✗
Xiang *et al.* [104]	5 min	Wind	ViT, LSTM	✓	✗	✓	✗	✗	✓	✗
Lai *et al.* [93]	10 min	Wind	BERT	✓	✗	✓	✗	✗	✗	✗
Yan *et al.* [102]	Hourly	Solar	GPT	✓	✗	✗	✓	✓	✗	✓
Wu *et al.* [97]	Hourly	Wind	GPT-2	✓	✗	✓	✓	✗	✗	✗
Akinci *et al.* [13]	Daily	Wind	DT, LLM	✓	✗	✗	✓	✗	✗	✓
Zhou *et al.* [99]	Hourly	Wind	LLM	✓	✓	✗	✗	✗	✗	✗
Zhu *et al.* [56]	15 min	Wind	LLM	✓	✗	✗	✓	✗	✓	✗
Liang *et al.* [100]	Hourly	Wind, Solar	GPT	✓	✗	✓	✗	✗	✗	✗
Fan *et al.* [45]	15 min	Wind	LLM	✓	✗	✓	✗	✗	✓	✗
Zhang *et al.* [91]	Weekly	Solar	LLaMA	✓	✗	✓	✓	✗	✗	✗
Fan *et al.* [95]	15 min	Wind	GPT	✓	✗	✗	✓	✗	✗	✗
Lin *et al.* [105]	1 min	Solar	VLM	✓	✗	✗	✓	✗	✓	✗
Fan *et al.* [96]	Hourly	Wind	LLM	✓	✓	✗	✓	✗	✗	✗
Chen *et al.* [103]	Hourly	Solar	LLM	✓	✓	✗	✗	✗	✗	✗
Duan *et al.* [98]	15 min	Wind	LLM	✓	✓	✗	✓	✗	✗	✗
He *et al.* [92]	Hourly	Solar	LLM	✓	✗	✗	✓	✗	✗	✗

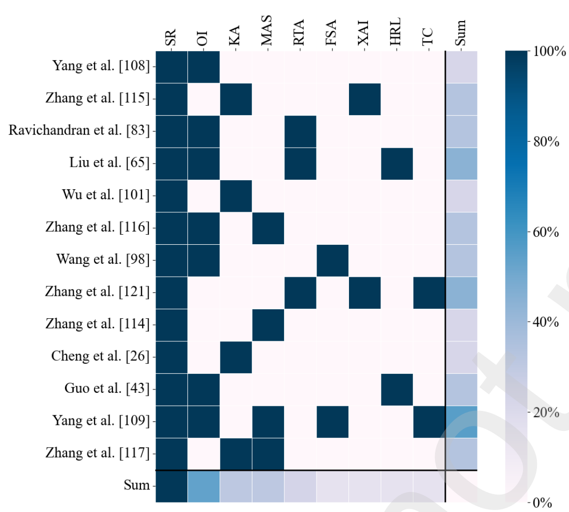

Fig. 5. Technical attribute distribution of LLM-based smart microgrid studies.

controllers cannot provide. The second direction integrates LLMs with DRL, as shown in Table 4, where LLMs assist in real-time decision-making, improving both decision quality and generalization abilities [108, 109, 110]. Furthermore, energy management-related knowledge is stored in external knowledge bases and integrated into the large model's input, thereby enhancing its domain expertise [58, 14, 9, 111]. Despite these advancements, LLMs still face challenges such as high inference latency, which hinders their ability to meet real-time control demands. Moreover, the security of decisions made by LLMs is not fully guaranteed, raising concerns about potential privacy breaches and security risks. A critical distinction here is the shift from "Black-box" to "White-box" intelligence. While DRL [108, 109] optimizes policies through trial-and-error, it offers little interpretability. In contrast, LLM-based EMS [107] can generate explanations for dispatch decisions, potentially increasing operator trust-a decisive factor for real-world deployment. As we explored in Section 3.3.5, future research could investigate MAS architectures where specialized agents handle distinct EMS functions with explicit coordination protocols to ensure safety and real-time responsiveness.

5.2. LLM-enhanced Control Systems

LLM-based agents enhance synchronized control systems by interpreting the system's operational status and using real-time data to maintain grid synchronization and stability [11, 106]. For instance, Zhang *et al.* [10] integrated multi-agent systems (MAS) with LLMs to detect and correct operational violations, combining semantic reasoning with numerical data. A power flow solver generates coordinated action sequences, while a sandbox rollback ensures system stability. Additionally, Guo *et al.* [112] proposed an LLM-based photovoltaic evaluation framework that optimizes microgrid control efficiency using few-shot learning, demonstrating the LLM's contextual learning ability in new scenarios. Traditional synchronization relies solely on electrical signals, whereas LLM agents can interpret system logs and operation manuals to understand the cause of desynchronization, enabling more resilient recovery strategies. Despite these advancements, synchronized control requires extremely high-speed responses, while MAS involves communication between multiple agents and reasoning processes within LLMs. Current methods struggle to meet the stringent real-time constraints needed for effective control. Furthermore, there is a lack of systematic design regarding the interaction between large models, human operators, and traditional control systems, particularly in terms of when control rights should be transferred between these entities. Future research could explore hierarchical control architectures that strategically partition responsibilities: fast inner-loop synchronization control handled by traditional model-based controllers, while LLM-based agents manage slower outer-loop coordination tasks.

Table 4
Large Language Model Approaches in Microgrids Smart Power Engineering

Ref.	Scenario	Base Model	Pre-trained	PEFT	SFT	ICL/Prompt	RAG	MLLM	API Call
Yang *et al.* [107]	Energy Management	LLM	✓	✗	✗	✓	✗	✗	✗
Zhang *et al.* [111]	Energy Management	LLM	✓	✗	✗	✓	✗	✗	✗
Ravichandran *et al.* [11]	Sync. Control	LLM, Agent	✓	✗	✗	✓	✗	✗	✗
Liu *et al.* [108]	Energy Management	LLM, SAC	✓	✗	✓	✗	✗	✗	✗
Wu *et al.* [58]	Energy Management	LLM, KG	✓	✗	✗	✗	✓	✗	✗
Zhang *et al.* [109]	Energy Management	LLM, Agent	✓	✗	✓	✗	✗	✗	✗
Wang *et al.* [106]	Sync. Control	LLM, Agent	✓	✗	✗	✓	✗	✗	✗
Zhang *et al.* [10]	Sync. Control	LLM, Agent	✓	✗	✗	✓	✗	✗	✗
Zhang *et al.* [14]	Energy Management	LLM	✓	✗	✗	✗	✓	✗	✗
Cheng *et al.* [9]	Energy Management	LLM	✓	✗	✓	✓	✗	✓	✗
Guo *et al.* [112]	Sync. Control	LLM	✓	✗	✗	✓	✗	✗	✗
Yang *et al.* [12]	Energy Management	LLM	✓	✗	✗	✓	✗	✗	✗
Zhang *et al.* [110]	Energy Management	GPT, DT	✓	✓	✗	✗	✗	✗	✗

Discussion on Future Directions: Microgrids with diverse topologies and heterogeneous energy configurations require customized control strategies. PEFT supports this by allowing each microgrid to maintain its own lightweight adapter. Furthermore, the power system is transitioning from centralized control to distributed collaboration. Multi-Agent Systems (MAS) equip each distributed energy resource (DER) with an intelligent agent that autonomously determines charging and discharging strategies. These agents coordinate their actions through negotiation, achieving local autonomy while optimizing for global outcomes. However, LLM inference latency is significantly higher than that of traditional controllers. For scenarios requiring rapid response, current MAS solutions cannot meet real-time requirements. A future solution involves cloud-edge collaboration systems, where the dispatch center deploys large models for advanced decision-making tasks and field terminals deploy smaller models for rapid response. Additionally, embodied intelligence can be implemented through robots and autonomous systems that understand language instructions, comprehend physical constraints, and execute operational tasks within complex power grid environments.

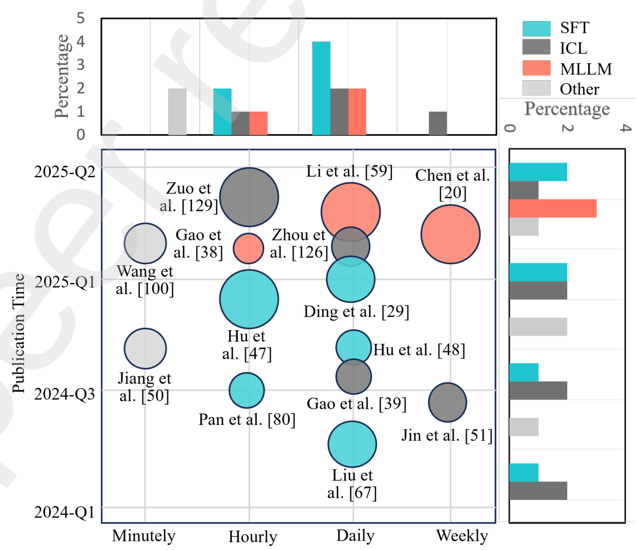

Fig. 6. Temporal and methodological distribution of LLM-based load forecasting research.

6. Load Management and Forecasting

Load management and load forecasting are key components of smart grid operation. The goal of load management is to achieve supply–demand equilibrium, reduce peak load, and enable effective demand response through proactive control of electrical equipment and optimization of energy dispatch strategies. This load management problem is typically modeled as a constrained optimization task. Load forecasting aims to estimate future electricity demand across different time periods, supporting operational scheduling, market participation, and capacity expansion planning. At its core, load forecasting is fundamentally a time-series prediction problem. Both domains face common challenges, such as high-dimensional state representation, uncertainty, interactions across different time scales, and heterogeneity in consumer behavior. Fig. 6 maps the reviewed LLM-based load-forecasting studies across prediction horizons and publication time. Most works concentrate on hourly and daily horizons, while only a few address minutely or weekly scales. In terms of methodology, fine-tuning-based approaches (SFT) and in-context learning (ICL) remain dominant, whereas multimodal LLMs (MLLM) and other hybrid strategies appear only in specific cases. Larger models tend to be used for longer horizons, reflecting the need to capture richer contextual dependencies. Current research works have progressed rapidly but remain clustered around a limited set of techniques, leaving ultra-short-term control and long-term planning comparatively underexplored [113, 114]. Therefore, this section examines how LLM-based techniques are applied to load management and forecasting, focusing on their capabilities in semantic understanding of operational contexts, few-shot adaptation to new scenarios, and multimodal information fusion to enhance prediction and control.

6.1. Load Management

Recently, LLMs have been applied to load management due to their strong semantic understanding. For example, time-series data can be converted into textual prompts, allowing LLMs to reason over temporal patterns and generate control strategies through natural language understanding [115]. However, since LLMs lack inherent domain-specific expertise in power systems, RAG frameworks have been introduced to integrate external knowledge bases, enabling interactive, knowledge-grounded decision support [116]. Moreover, given the substantial energy consumption of GPU model training and inference, several LLM scheduling frameworks have been proposed for heterogeneous GPU clusters, aiming to reduce power usage and improve energy efficiency in resource-constrained environments [113, 17]. Despite these advantages, the high inference latency of LLMs makes them unsuitable for millisecond-level real-time control, and their decisions lack explicit guarantees for constraints, potentially violating physical or safety limits. This latency-safety trade-off implies that LLMs are currently best suited for "Outer-Loop" planning tasks (e.g., day-ahead scheduling negotiation) rather than "Inner-Loop" control (e.g., frequency regulation). The cognitive reasoning of LLMs should complement, not replace, the millisecond-level reflex of PID or MPC controllers. Additionally, as the grid topology evolves or new devices are integrated, efficient and low-cost model updates remain an open research challenge. Hybrid architectures could represent a promising pathway forward, which strategically partition control responsibilities between fast traditional controllers and slower LLM-based planning agents.

6.2. Load Forecasting

Load forecasting aims to predict future electricity demand across various timescales based on historical load data. With the advent of LLMs, Transformer-based pre-trained architectures have been increasingly applied to load forecasting tasks, as shown in Table 5, leveraging self-attention mechanisms to capture long-range dependencies in time-series data [117, 118, 119]. Directly updating all parameters of these large pre-trained models is computationally expensive. Therefore, PEFT techniques have been introduced to efficiently adapt LLMs to power system forecasting by updating only a small subset of parameters [120, 121]. Given that load forecasting is influenced by numerous external factors, MLLMs have been employed to integrate heterogeneous data sources such as news reports, policy documents, and weather forecasts, thereby improving predictive performance and robustness [59, 122, 123, 116, 124, 114]. A core advantage of LLMs lies in their ability to process unstructured information, such as sudden policy changes, public holiday shifts, or social events, which traditional time-series models often treat as noise or require manual feature engineering to capture.LLMs naturally ingest these textual contexts to calibrate numerical forecasts. Furthermore, multi-agent LLM architectures have extended modeling performance by introducing collaborative reasoning mechanisms. For instance, Zuo *et al.* [15] proposed a multi-agent framework that embeds interactive processes throughout the forecasting pipeline, enhancing inference capabilities through coordinated agent collaboration. However, the modality gap between time-series and textual data can cause misalignment issues, as LLM tokenization mechanisms may incorrectly encode floating-point values, impairing the model's ability to learn precise numerical relationships. Inadequate semantic alignment between modalities may also degrade performance. Critically, existing LLM-based approaches lack an inherent understanding of the physical constraints governing power systems, which can lead to forecasts that violate physical laws, limiting their trustworthiness and applicability in real-world operations.

Discussion on Future Directions: Load management involves complex decision-making based on diverse data sources. MLLMs can integrate real-time load curves, market transaction data, and news sentiment analysis to generate more informed dispatch schemes. LLMs transform load forecasting from curve-fitting into cognitive interpretation of policy changes, social behavior, and market dynamics. Additionally, power system-related technical standards are frequently updated, and RAG allows for the seamless integration of the latest versions of these standards into the model's knowledge base. Looking towards future simulation capabilities, the development of World Models allows AI systems to predict future states and enhance decision-making. In power systems, a world model acts as a digital twin of the grid, simulating both the physical laws governing the grid and learning from stochastic processes and human decision-making patterns, offering more flexible simulation capabilities.

7. Agent Systems in Power Markets

In electricity market trading, an agent is a computational entity with autonomous decision-making capabilities. It can perceive the market environment, formulate strategies, and execute energy transactions. Market participants can operate as independent entities, enabling peer-to-peer (P2P) or community-level energy trading without the need for a central coordinator. In P2P trading, prosumers trade electricity directly, bypassing intermediaries like grid operators. Community energy trading involves nearby users coordinating their energy production, storage, and consumption to maximize welfare and achieve self-sufficiency. Both scenarios can be formulated as distributed optimization problems, where agents must make optimal decisions, given challenges such as dynamic pricing, network constraints, and strategic interactions. Before the rise of LLMs, machine learning, deep reinforcement learning, and federated learning were commonly applied to these prediction problems. With the advent of LLMs, their natural language understanding capabilities allow agents to better comprehend market rules, improving decision-making, where LLMs most clearly evolve from predictors into cognitive strategists.

Table 5
Large Language Model Approaches for Load Management and Forecasting

Ref.	Scenario	Base Model	Pre-trained	PEFT	SFT	ICL/Prompt	RAG	MLLM	API Call
Pan *et al.* [117]	Load Forecasting	Transformer	✓	✗	✓	✗	✗	✗	✗
Sun *et al.* [118]	Load Forecasting	Informer	✓	✗	✓	✗	✗	✗	✗
Wang *et al.* [113]	Energy Management	LLM	✓	✗	✗	✗	✗	✗	✗
Zhou *et al.* [122]	Load Forecasting	LLM	✓	✗	✗	✓	✗	✗	✗
Li *et al.* [59]	Load Forecasting	GPT	✓	✗	✗	✗	✗	✓	✗
Liu *et al.* [120]	Load Forecasting	LLM	✓	✓	✗	✓	✗	✗	✗
Zuo *et al.* [15]	Load Forecasting	LLM, Agent	✓	✗	✗	✓	✗	✗	✗
Ding *et al.* [119]	Load Forecasting	LLM, MoE	✓	✗	✓	✗	✗	✗	✗
Jin *et al.* [115]	Energy Management	LLM	✓	✗	✗	✓	✗	✗	✗
Jiang *et al.* [17]	Energy Management	LLM	✓	✗	✗	✗	✗	✗	✗
Chen *et al.* [116]	Load Forecasting	LLM	✓	✗	✗	✓	✓	✓	✗
Hu *et al.* [123]	Load Forecasting	BERT	✓	✗	✓	✗	✗	✗	✗
Gao *et al.* [114]	Load Forecasting	GPT	✓	✗	✓	✗	✗	✓	✗
Hu *et al.* [121]	Load Forecasting	GPT	✓	✓	✗	✗	✗	✗	✓
Gao *et al.* [124]	Load Forecasting	LLM	✓	✗	✗	✓	✗	✗	✗

Fig. 7 shows the temporal evolution of technical capabilities in LLM-based power market research. The horizontal bars reflect the capability adoption rates, while the vertical bars indicate quarterly publication counts. We find that Autonomous Decision (AD) is consistently present in all studies, indicating its central role in power market research. The adoption of Multi-Agent System (MAS) is high (83%) and remains steady, while Market Semantic Reasoning (MSR) and Knowledge Augmentation (KA) see a gradual increase in their use, reaching 50% adoption. In terms of publication trends, the most notable peak occurs in 2024-Q2, when 50% of studies adopt multiple technical capabilities simultaneously. Behavioral Strategy Modeling (BSM) and Human-like Negotiation (HN) remain less frequent but show some emergence toward the end of the timeline. These observations highlight that while AD continues to dominate, newer techniques like MAS [125, 126], BSM [127], and HN [128] are beginning to gain traction, suggesting a shift towards more dynamic and interactive models for power market applications. This section explores how LLM-based techniques are transforming P2P and community energy trading, focusing on their ability to reason semantically about market mechanisms, adapt quickly to regulatory changes, and enable human-like strategic interactions for intelligent market participation.

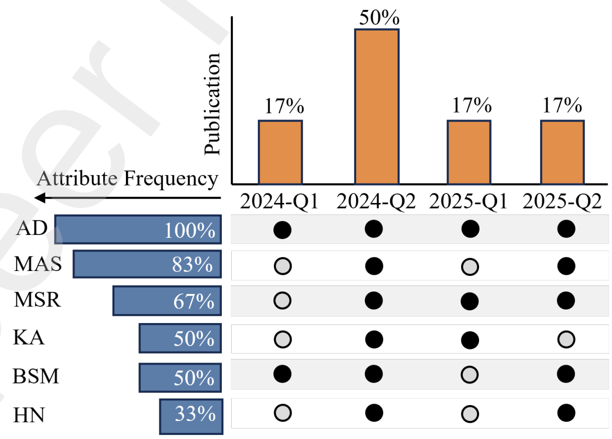

Fig. 7. Temporal evolution of technical capabilities in LLM-based power market research.

7.1. P2P Energy Trading

LLMs enhance decision-making in P2P energy trading by introducing advanced levels of intelligence, as shown in Table 6. For example, Chen et al. [125] introduced an interactive learning framework that combines ChatGPT with reinforcement learning to support automated, interactive energy trading. This framework enables agents to understand trading intentions through natural language and generate effective strategies. Lou et al. [126] employed LLMs as expert systems to generate personalized trading strategies and guide multi-agent imitation learning, helping new agents absorb expert knowledge without relying on large volumes of historical data. These approaches leverage the semantic understanding and reasoning capabilities of LLMs, allowing agents to comprehend market rules and constraints while rapidly adapting to new trading scenarios through few-shot learning with minimal examples. Unlike traditional game theory approaches [129] that assume perfect rationality, LLM-based agents [125, 126] can simulate human-like negotiation behaviors. This capability allows for more realistic market simulations that account for psychological factors, trust building, and imperfect information, which are prevalent in distributed P2P markets. However, LLMs face challenges such as high inference latency, making them unsuitable for real-time trading. Additionally, trading strategies generated by LLMs may conflict with grid constraints and lack sufficient built-in safety verification mechanisms. Furthermore, changes in market rules or trading structures require retraining or fine-tuning of LLMs, which is both resource-intensive and time-consuming. As we elaborated in Section 3.3.5, MAS could coordinate specialized trading agents with distinct roles, utilizing explicit communication protocols and safety verification layers before executing transactions. Federated LLM frameworks can also enable collaborative learning of trading strategies across multiple prosumers while preserving transactional privacy and strategic confidentiality, addressing data scarcity challenges in emerging P2P markets with limited historical data.

7.2. Community Energy Trading

By leveraging semantic understanding, knowledge reasoning, and the ability to integrate multimodal information, LLMs introduce a higher level of decision support and market analysis into community energy trading. For example, Lu et al. [127] proposed fine-tuned LLMs to create bidding behavior agents, analyze market sentiment, and generate conditional time series through GANs, enabling communities to adjust trading strategies based on market dynamics. Kar et al. [130] developed agents that retrieve weather and economic indicators to generate interpretable reports for energy market analysis, improving the transparency and credibility of decision-making. Zhang et al. [111] transformed battery storage bidding into a DRL problem, enhancing risk management through Conditional Value-at-Risk. This approach enables community storage systems to make robust decisions under uncertainty. Tian et al. [128] integrated hierarchical memory to simulate human-like decision-making and model delays in market behavior, enabling LLMs to capture the long-term strategic intentions of market participants. In fact, community trading often involves complex social norms, such as fairness and empathy, which are difficult to formulate mathematically in standard objective functions. LLMs enable agents to interpret and adhere to these qualitative community rules through natural language prompts, bridging the gap between social values and economic optimization. However, community energy trading involves multiple objectives, such as cost, emissions, and fairness, and LLMs may not fully address all these goals simultaneously. Additionally, there is a lack of profit distribution mechanisms using LLMs that incentivizes individual participation while ensuring overall community fairness. Each community member could be represented by a personalized LLM agent that negotiates on their behalf, while a community-level coordinator agent ensures that collective objectives and fairness constraints are met.

Discussion on Future Directions: In the diversified power market, MAS can simulate more realistic market behavior, aiding in the formulation of regulatory policies. However, power system collaborative optimization involves multiple conflicting objectives, including economic efficiency, safety, environmental protection, and fairness. One approach is to introduce a social welfare term, where an agent's reward includes not only its own gains but also its contribution to global welfare. Data privacy remains a critical issue. Federated Large Language Modeling (FLLM) offers a privacy-preserving distributed training approach. In power market strategy optimization, various market participants can jointly train a market forecasting model, sharing insights into market dynamics without disclosing their respective bidding strategies and cost structures. To ensure sustainable operation, game theory methods, such as Shapley values, should be used to quantify each participant's contribution. Differentiated incentives should be designed to reward data providers who contribute high-quality information.

8. Security and Safety Challenges Associated with Large Language Models

The integration of LLMs into power systems represents a paradigm shift in grid protection. Unlike conventional ML models that focus on numerical anomaly detection, LLMs introduce semantic reasoning capabilities, allowing the grid to interpret unstructured data such as system logs, maintenance reports, and cyber-threat intelligence. However, this transition also introduces unique risks. LLMs serve as a double-edged sword: they function as intelligent guardians capable of reasoning about complex threats, yet they also introduce new attack surfaces like prompt injection and intrinsic reliability issues such as hallucinations. This section explores this duality, focusing strictly on the security landscape reshaped by Large Models.

8.1. LLM-Enhanced Defensive Mechanisms

Traditional grid defenses rely on numerical algorithms to detect statistical deviations in sensor data [131, 132]. While effective against known patterns, these methods lack the ability to understand the context of an attack. LLMs fill this gap by evolving defense from numerical detection to semantic threat intelligence.

A primary advantage of LLMs lies in automated log analysis and forensics. Power grids generate vast amounts of unstructured log data that traditional models struggle to process. LLMs can parse these logs in real-time to identify subtle attack sequences that numerical models miss. For instance, Zaboli et al. [133] demonstrated that LLMs can analyze substation network traffic logs to detect multi-stage cyber-attacks by reasoning about the correlation between seemingly unrelated events. Furthermore, LLMs significantly enhance the timeliness of operational safety verification through RAG. Power system operation regulations are frequently updated to reflect new equipment integration or safety standards, and traditional models typically require retraining to adapt to these changes. In contrast, RAG enables LLMs to access dynamic external knowledge bases without model updates. Cheng et al. [134] proposed a framework where LLM agents retrieve the most recent dispatch operation regulations to automatically verify operator commands in real-time. This approach ensures that safety checks are always aligned with the latest grid codes, significantly reducing the risk of non-compliant operations caused by outdated knowledge.

Beyond log and regulation analysis, LLMs can also detect anomalies in critical control instructions that traditional numerical classifiers might overlook. Selim et al. [135] introduced a framework that converts numerical Volt/VAR

Table 6
Large Language Model Approaches for Decentralized Energy Trading: Taxonomy of Key Techniques

Ref.	Scenario	Base Model	Pre-trained	PEFT	SFT	ICL/Prompt	RAG	MLLM	API Call
Lu et al. [127]	Community	LLM, GAN	✓	✗	✓	✗	✗	✓	✗
Kar et al. [130]	Community	LLM, Agent	✓	✗	✗	✓	✓	✗	✓
Zhang et al. [111]	Community	LLM, Agent	✓	✗	✓	✗	✗	✗	✗
Tian et al. [128]	Community	LLM	✓	✗	✗	✓	✓	✗	✗
Chen et al. [125]	P2P	GPT, Agent	✓	✗	✗	✓	✗	✗	✓
Lou et al. [126]	P2P	LLM, Agent	✓	✗	✓	✓	✗	✗	✗

control commands of smart inverters into natural language descriptions. By leveraging the semantic understanding of LLMs, this method can identify cyber-attacks targeting voltage regulation curves, acting as a cognitive firewall for device-level control interfaces.

8.2. Vulnerabilities of Large Models

While LLMs enhance defense, they introduce a new class of generative vulnerabilities that differ fundamentally from traditional software bugs or adversarial examples in Deep Learning. The most prominent risk stems from the natural language interface itself, which creates vectors for prompt injection and data leakage. Unlike numerical inputs, linguistic inputs are difficult to sanitize. An attacker could craft a malicious prompt-disguised, for example, as a maintenance command-to manipulate the LLM-based scheduler into ignoring safety constraints or revealing sensitive grid topology [136]. In addition to interaction-based attacks, data privacy presents another critical challenge. Since LLMs are trained on massive datasets, they may inadvertently memorize sensitive power grid information, such as user load profiles or critical infrastructure maps. Research indicates that LLMs are susceptible to data extraction attacks, where adversaries query the model to reconstruct private training data [137]. Research highlights that using raw network data for model training raises significant privacy concerns, necessitating the use of privacy-preserving encoding techniques to prevent the model from memorizing and inadvertently leaking critical infrastructure details [138].

8.3. Intrinsic Safety and Operational Prudence

Another challenge for applying LLMs in power systems is their intrinsic operational safety. A fundamental conflict exists between the nature of power grids and LLMs. Power systems operate under strict, deterministic physical laws (e.g., Kirchhoff's laws, thermal limits) with zero tolerance for violations. In contrast, LLMs are probabilistic engines based on next-token prediction. This mismatch leads to the risk of hallucinations, where an LLM might confidently generate a dispatch schedule that is linguistically fluent but physically infeasible or dangerously unstable. Given these risks, relying solely on the reasoning of LLMs is insufficient for critical grid control. Machlev et al. [36] highlighted the dangers of "black-box" decision-making in high-stakes environments. Therefore, LLMs should function strictly as advisory copilots, assisting operators with information synthesis and draft generation. Their outputs must pass through rigorous “physics-informed guardrails"-such as deterministic numerical solvers and Power Flow verification-before any command is executed on the physical grid.

9. Conclusion

In this paper, we surveyed over 129 technical papers on artificial intelligence applications in power systems, covering machine learning, deep learning, and large language models. LLMs mark the transition from AI as a forecasting tool to AI as a cognitive operator - capable of reasoning over rules, coordinating tools, collaborating with humans, and ensuring safe, constraint-aware operation. However, numerous open challenges remain, including the integration of physical constraints into LLM architectures, mitigating hallucinations in safety-critical grid operations, and developing efficient real-time inference mechanisms for control applications. Achieving intelligent and trustworthy power systems will require collective efforts from the global research community, industry practitioners, and policymakers. We hope this survey serves as a valuable resource for researchers and practitioners, driving ongoing efforts to build safe, robust, and intelligent next-generation power systems in the era of large language models.

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